

# ■ UZHAVU KAAPPAAN (■■■■■ ■■■■■■■■■■)

Smart Crop Rotation & Farmer Soil Feeding Action Plan  
Data Mode: ■ Verified Lab Ingestion (Device: soil-scout-02) · Certified P025 Model

Farmer: Farmer #102 | Land Area: 4.5 Acres  
Location: Farm #102 (11.0168° N, 76.9558° E) | Irrigation: Drip  
Plan Ref ID: #UK-P025-FARM-102 | Report Date: 03 Oct 2026, 12:25 AM IST  
Live Reading: Active & Calibrated

## 1. Live Sensor Diagnostic & Soil Fertility Telemetry

| ■■ Sunlight / Light        | ■ Soil Moisture             | ■■ Temperature               | ■ TDS Minerals                  | ■ Probe Reliability         |
|----------------------------|-----------------------------|------------------------------|---------------------------------|-----------------------------|
| 74%<br>Optimal photoperiod | 45%<br>Rhizosphere moisture | 31.5 °C<br>Soil probe sensor | 420 ppm<br>Conductivity & salts | ■ Reliable<br>Soil Scout v2 |

| Nutrient Parameter  | Real-Time Value | Target Range   | Deficit / Status      | Agronomic Impact & Crop Response                        |
|---------------------|-----------------|----------------|-----------------------|---------------------------------------------------------|
| Nitrogen (N)        | 65.0 kg/ha      | 80 - 160 kg/ha | -15.0 kg/ha (Deficit) | Restricts leaf canopy; urgent pulse rotation needed     |
| Phosphorus (P)      | 35.0 kg/ha      | 30 - 60 kg/ha  | Sufficient            | Affects root elongation & early plant establishment     |
| Potassium (K)       | 72.0 kg/ha      | 60 - 120 kg/ha | Sufficient            | Adequate for disease resistance, cell turgor & pod fill |
| Soil pH             | 7.20            | 6.0 - 7.5      | Neutral / Optimal     | Governs cation exchange & micronutrient availability    |
| Organic Carbon (OC) | 0.68%           | 0.80 - 1.50%   | -0.12% Deficit        | Governs moisture holding capacity & soil microbiome     |

## 2. Prescribed Soil Feeding & Fertilizer Dosage Schedule (for 4.5 Acres Land)

| Feeding Stage                        | Recommended Inputs                                                | Dose / Acre                         | Total (4.5 Ac)                    | Application Method & Agronomic Objective                                                                  |
|--------------------------------------|-------------------------------------------------------------------|-------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------|
| Basal Land Prep<br>(Prior to Sowing) | FYM / Vermicompost<br>Neem Cake<br>Bio-fertilizer (Rhizobium+PSB) | 4.0 Tonnes<br>100 kg<br>2 kg + 2 kg | 18.0 Tonnes<br>450 kg<br>18.0 kg  | Incorporate during final ploughing to boost Organic Carbon & inoculate root nodules                       |
| Sowing Time<br>(Basal Starter)       | DAP (Di-Ammonium Phos.)<br>MOP (Potash)<br>Zinc Sulphate (ZnSO4)  | 25 kg<br>15 kg<br>10 kg             | 112.5 kg<br>67.5 kg<br>45.0 kg    | Apply in bands 5cm below seed line for rapid root growth & zinc deficiency correction                     |
| Vegetative Growth<br>(25-30 Days)    | Urea (Top Dressing)                                               | 15 kg (-40% saved)                  | 67.5 kg<br>Save 45 kg (Rs. 1,575) | Groundnut biologically fixes nitrogen; saves 45 kg urea (Rs. 1,575) across 4.5 acres and prevents lodging |
| Flowering & Pods<br>(45-50 Days)     | 19:19:19 Soluble NPK<br>Borax (0.2% Boron)                        | 1.0 kg (Foliar)<br>200 g (Foliar)   | 4.5 kg<br>0.90 kg                 | Early morning foliar spray to prevent flower drop and ensure full pod filling                             |

## 3. Agronomic Risk Assessment & Recommended Restorative Crop Rotation

| Season 1 (Immediate Restoration)                                                                                                                                                                                                                                                                          | Season 2 (Soil Building)                                                                                                                                                                                                                                                               | Season 3 (Biomass & Stabilization)                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Groundnut (Top Match: 88.5%)</b> <ul style="list-style-type: none"><li>• Crop Family: Legume</li><li>• Duration: 65-85 Days</li><li>• Est. Net Profit: Rs. 29,000 / acre</li><li>• Biological Role: Biologically fixes 35-45 kg N/ha naturally, cuts synthetic urea, and breaks pest cycles.</li></ul> | <b>Guar seed</b> <ul style="list-style-type: none"><li>• Crop Family: Legume</li><li>• Duration: 65-85 Days</li><li>• Est. Net Profit: Rs. 15,000 / acre</li><li>• Biological Role: Biologically fixes 35-45 kg N/ha naturally, cuts synthetic urea, and breaks pest cycles.</li></ul> | <b>Green Gram</b> <ul style="list-style-type: none"><li>• Crop Family: Legume</li><li>• Duration: 65-85 Days</li><li>• Est. Net Profit: Rs. 6,482 / acre</li><li>• Biological Role: Biologically fixes 35-45 kg N/ha naturally, cuts synthetic urea, and breaks pest cycles.</li></ul> |

## 4. Multi-Season Financial Returns & Soil Health Recovery Trajectory

| Rotation Stage    | Crop Cultivated      | Cost / Ac  | Gross / Ac | Net Profit / Ac | Farm Total (4.5 Ac) | Soil Trajectory     |
|-------------------|----------------------|------------|------------|-----------------|---------------------|---------------------|
| Baseline          | Depleted State       | Rs. 36,000 | Rs. 48,000 | Rs. 12,000      | Rs. 54,000          | 74 / 100 (Current)  |
| Season 1 (Kharif) | Groundnut            | Rs. 18,000 | Rs. 33,700 | Rs. 29,000      | Rs. 130,500         | 78 / 100 (+4 pts)   |
| Season 2 (Rabi)   | Guar seed            | Rs. 16,000 | Rs. 31,000 | Rs. 15,000      | Rs. 67,500          | 81 / 100 (+3 pts)   |
| Season 3 (Zaid)   | Green Gram           | Rs. 11,000 | Rs. 17,482 | Rs. 6,482       | Rs. 29,169          | 84 / 100 (Restored) |
| 3-Season Total    | Restorative Rotation | Rs. 45,000 | Rs. 82,182 | Rs. 50,482 / ac | Rs. 227,169 Total   | 84 / 100 (Restored) |

- **Mandatory Agronomic Field Guidelines for Farmer #102:**
1. **Bio-Inoculation:** Treat Groundnut seeds with *Rhizobium* bio-fertilizer @ 25g/kg seed to maximize root nodulation.
  2. **Input Cost Savings:** Restricting top-dressed urea to 15 kg/acre saves 45 kg chemical urea (Rs. 1,575 direct savings across 4.5 acres) thanks to legume atmospheric N fixation.
  3. **Crop Residue Retention:** Do not burn crop stubbles. Plough haulms back into soil to lift Organic Carbon from 0.68% towards 0.80%.
  4. **Precision IoT Monitoring:** Keep Soil Scout probe clean; observe moisture reading (45%) to trigger irrigation when soil reaches 35%.

Certified by: UZHAVU KAAPPAAN P025 Model  
Telemetry: Live Soil Scout Probe #soil-scout-02 · 45,000+ Agricultural Training Records

Authorized Agronomist Extension Officer:

Department of Agriculture & Precision Agronomy, Farm #102